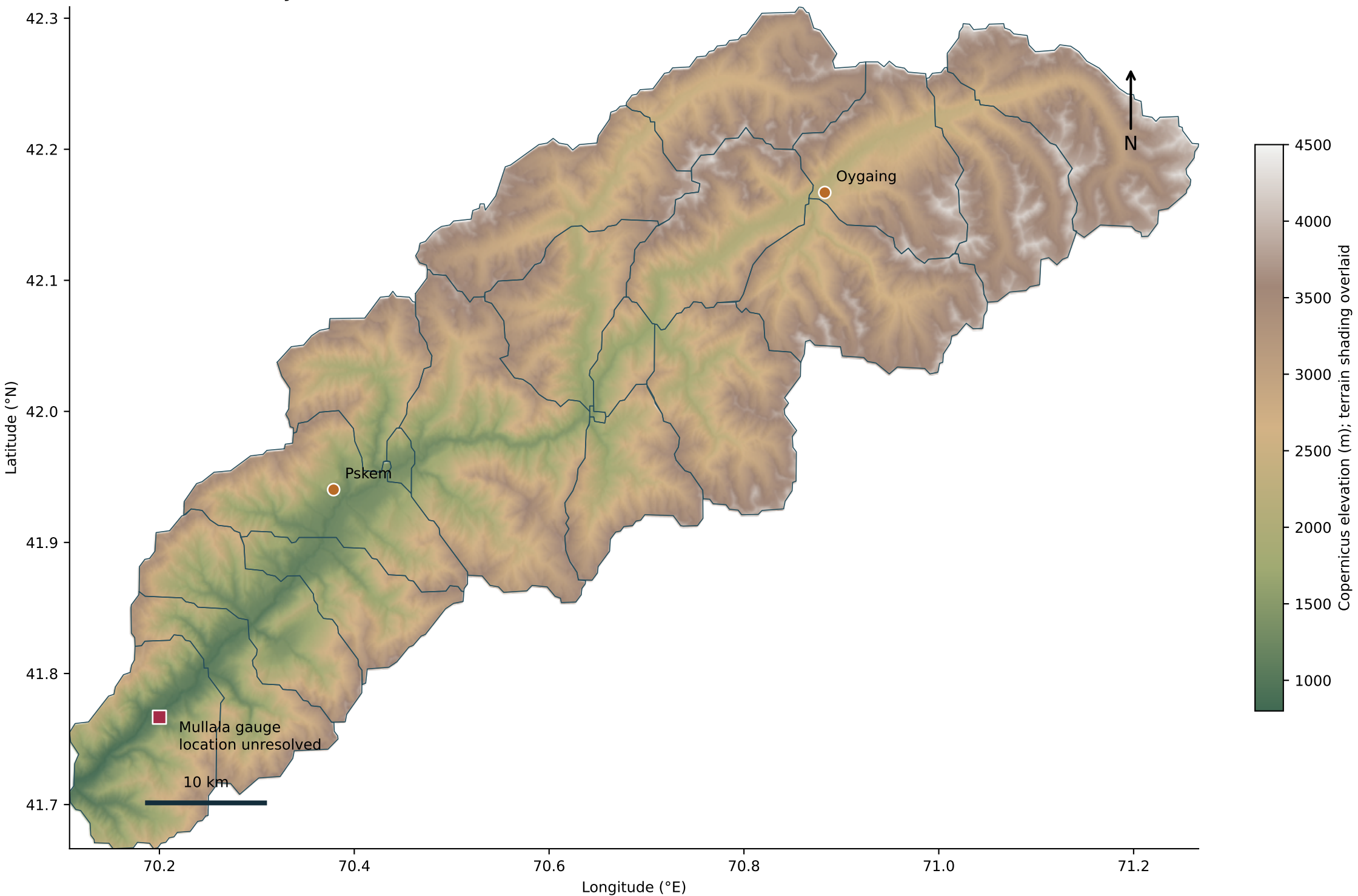


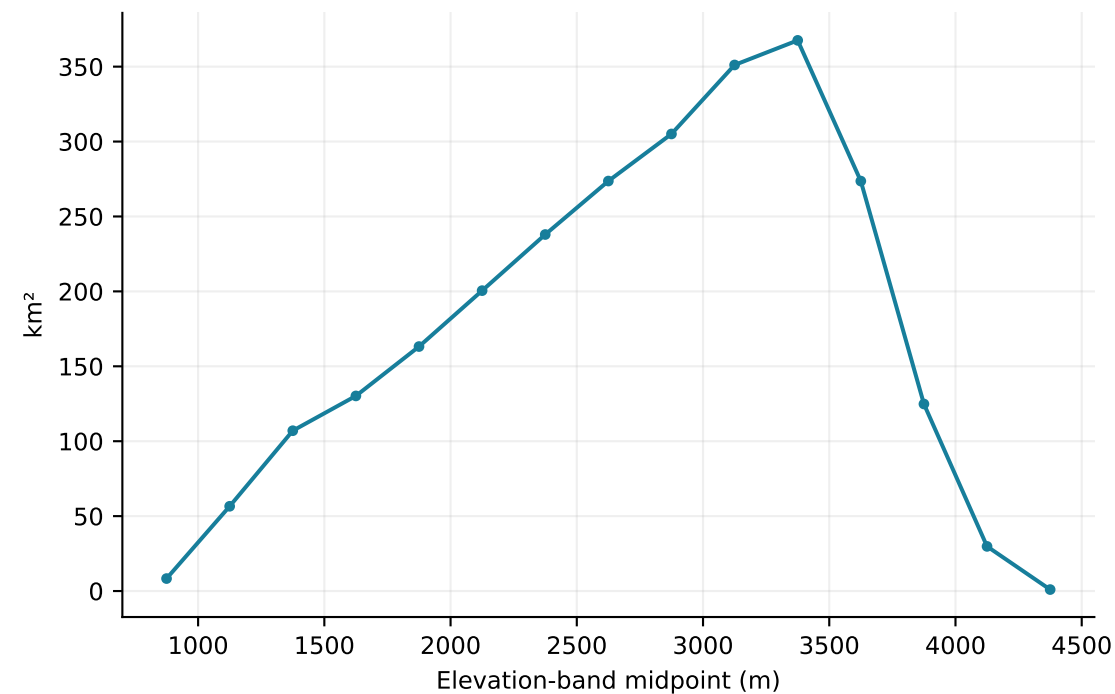
Pskem mountain study domain and observation locations



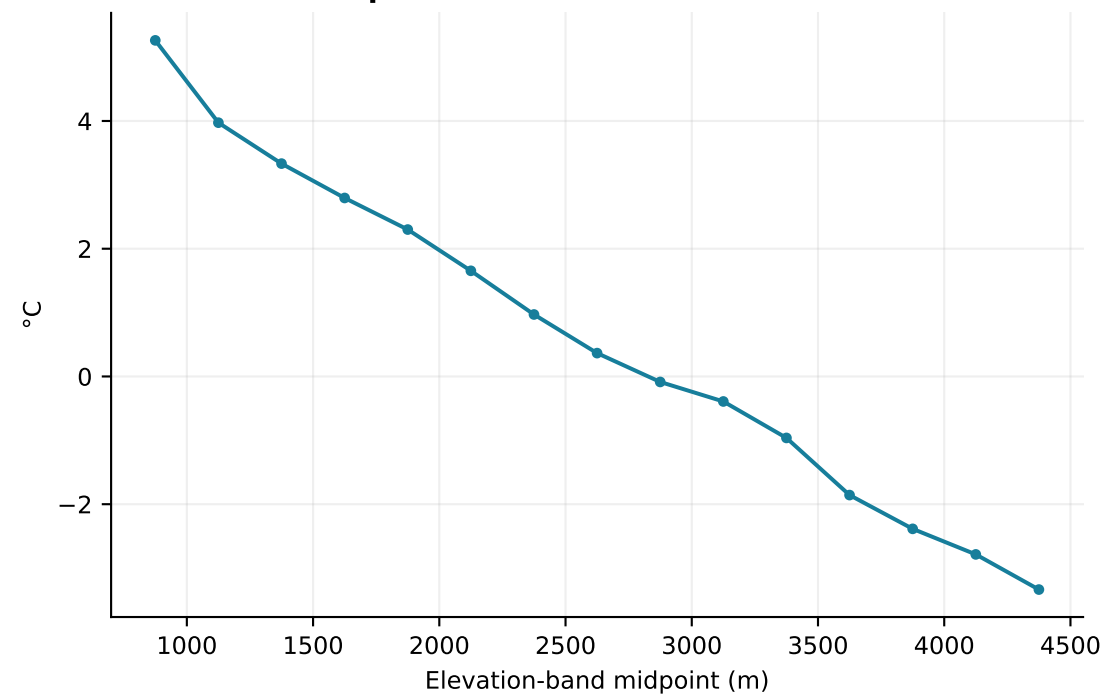
Sources: Copernicus DEM GLO-30 2024_1; supplied station coordinates; HydroBASINS level-12 trace.
The boundary includes the complete outlet unit and is provisional. Tashkent lies outside this mountain map.

Pskem: terrain and climate by 250 m elevation band

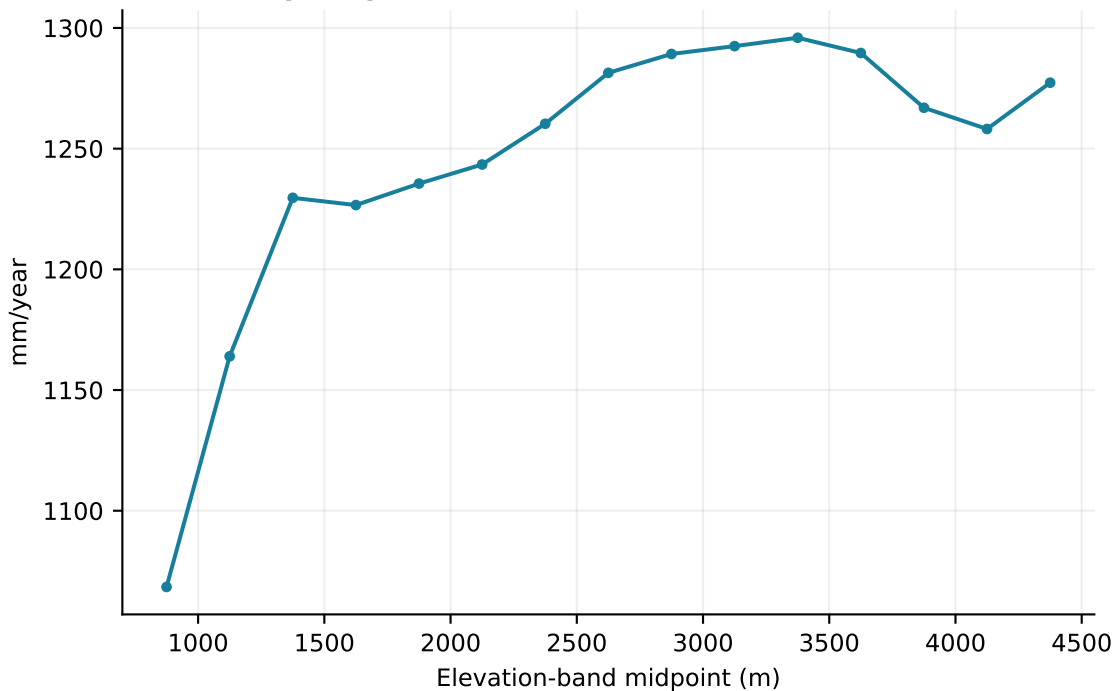
a. Mountain area



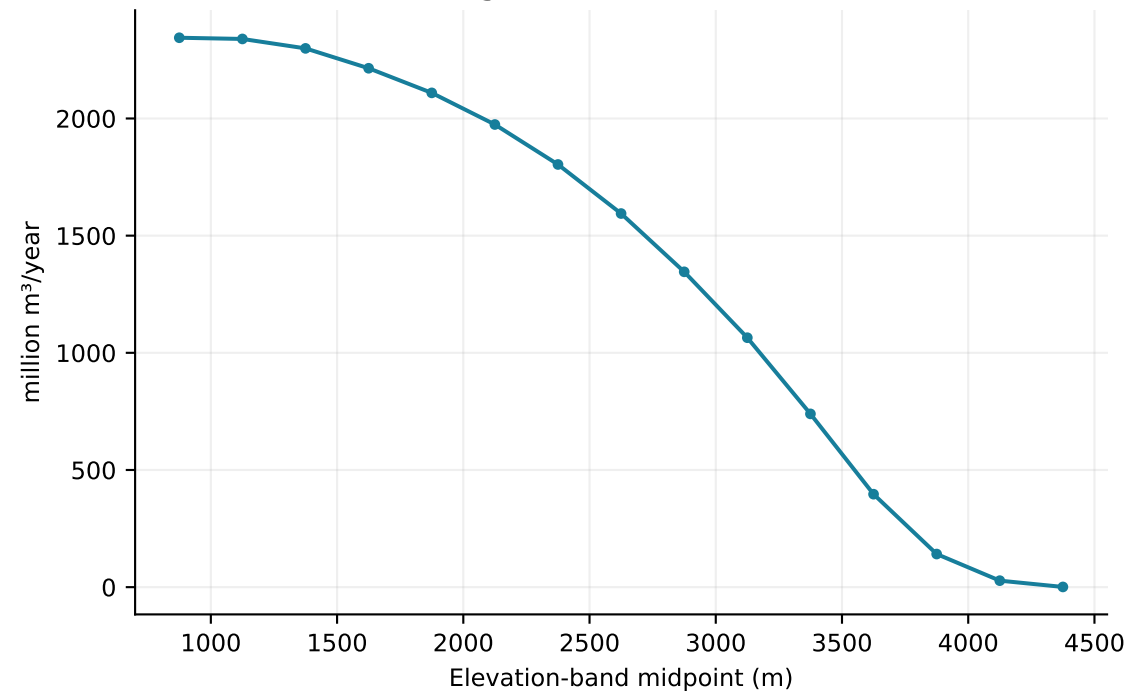
b. Mean air temperature



c. Annual precipitation

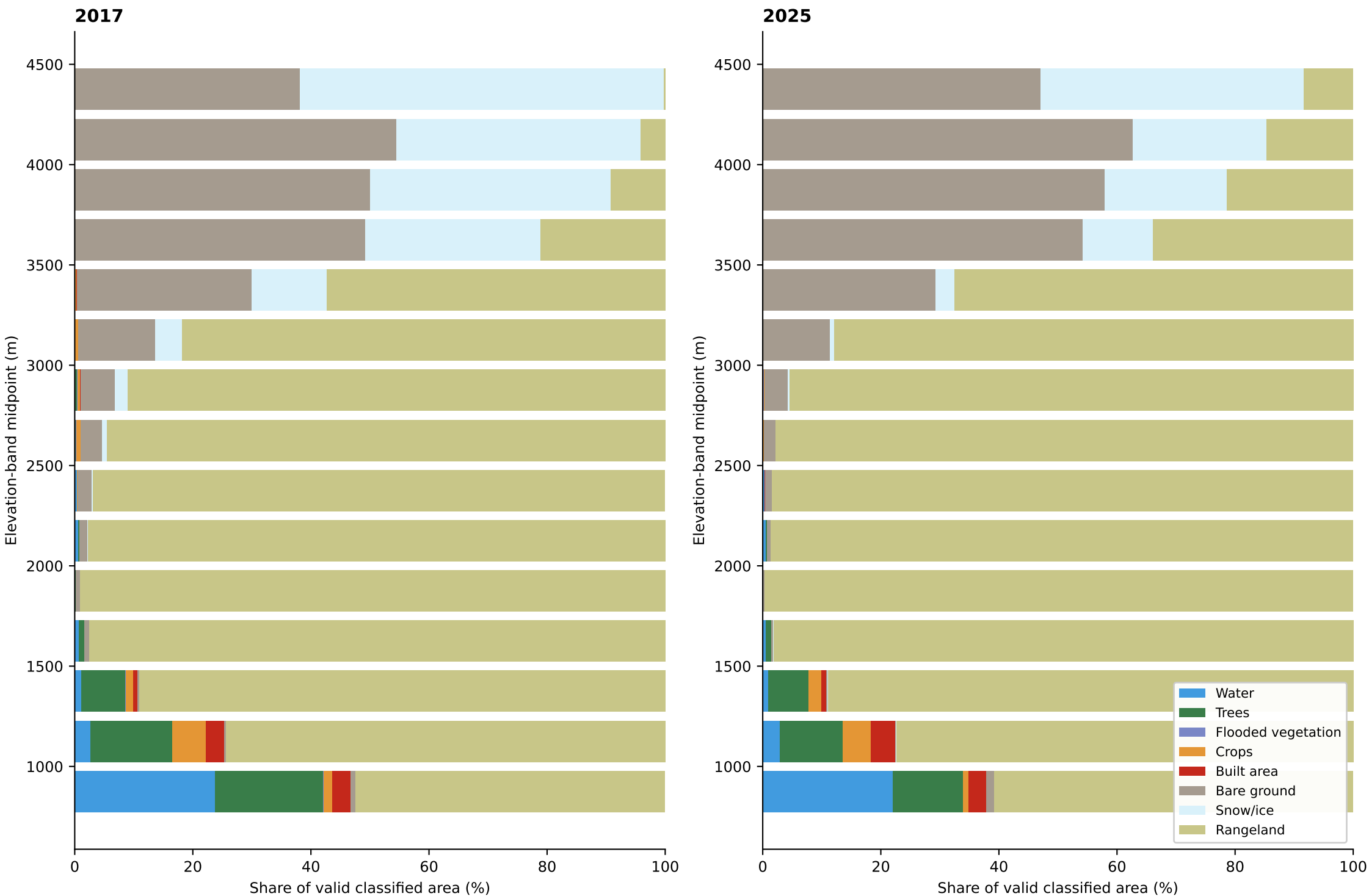


d. Cumulative runoff generated above



Sources: Copernicus DEM 30 m; ERA5-Land 2010-2024. Coarse climate sampled across terrain bands; not fine-scale downscaling.
Cumulative runoff is modelled generation, not routed or measured discharge. Provisional catchment.

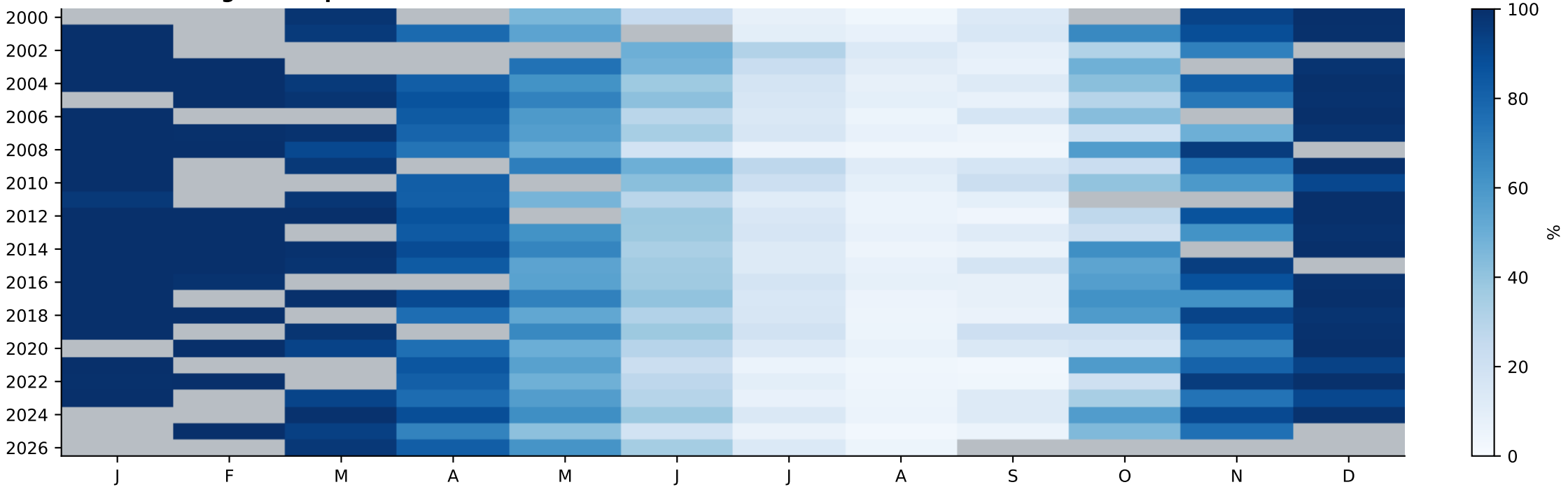
Pskem: 10 m land cover along the mountain



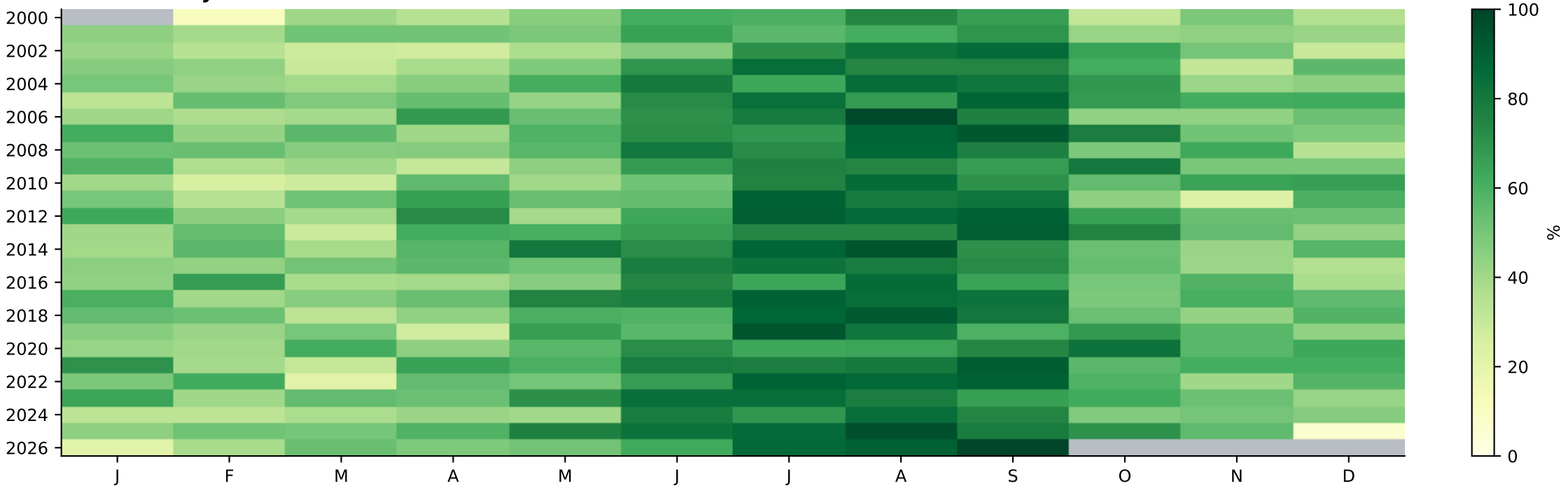
Source: Esri / Impact Observatory / Microsoft, community-hosted Earth Engine series.
Changes combine real change and classification differences. No local independent accuracy sample is available.

MODSNOW-style monitoring: snow and observation completeness

a. Snow among usable pixels



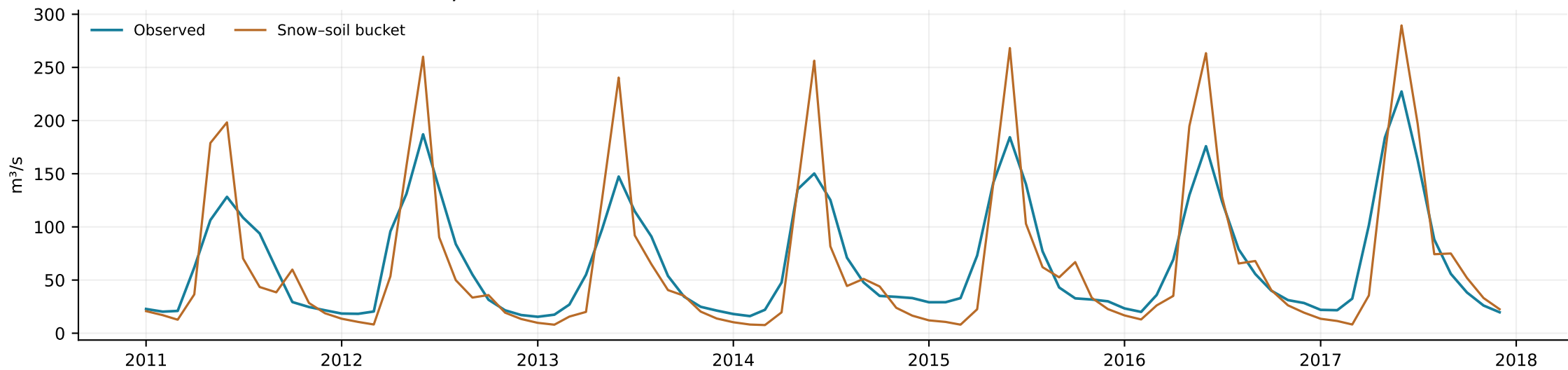
b. Mean daily usable area



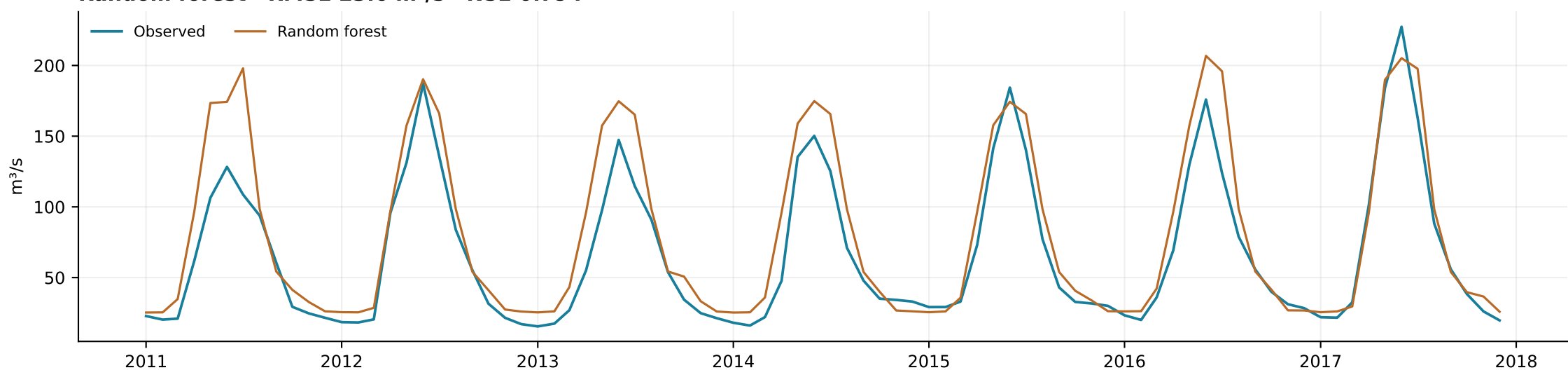
Sources: MOD10A1 + MYD10A1 v6.1. Same-day Terra priority, Aqua fallback; no temporal interpolation.
Snow means need $\geq 70\%$ daily valid area and ≥ 10 qualifying days/month. Grey means missing or insufficient observations.

Monthly model verification on unseen 2011-2017 observations

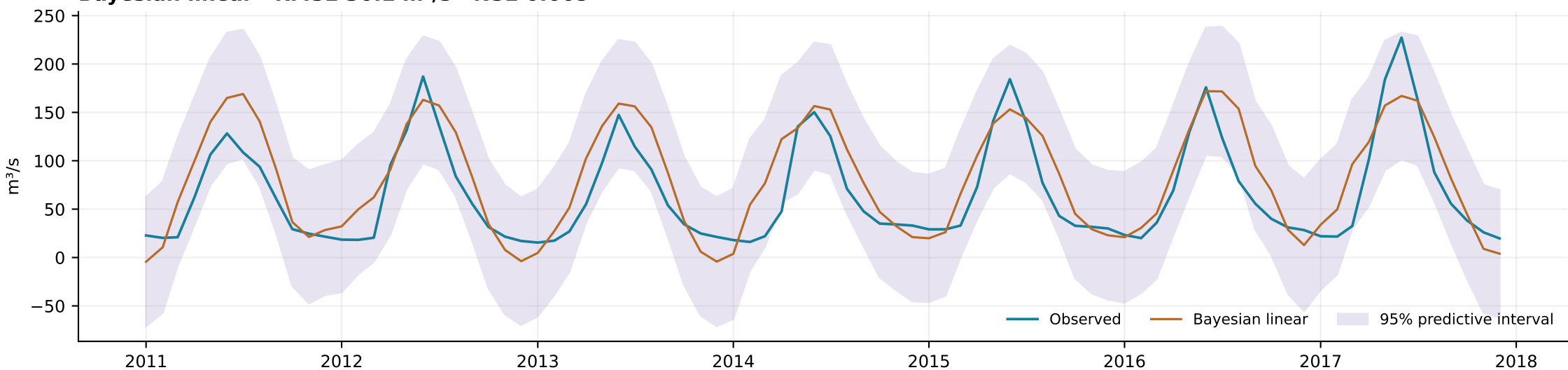
Snow-soil bucket • RMSE 33.6 m³/s • NSE 0.582



Random forest • RMSE 23.6 m³/s • NSE 0.794



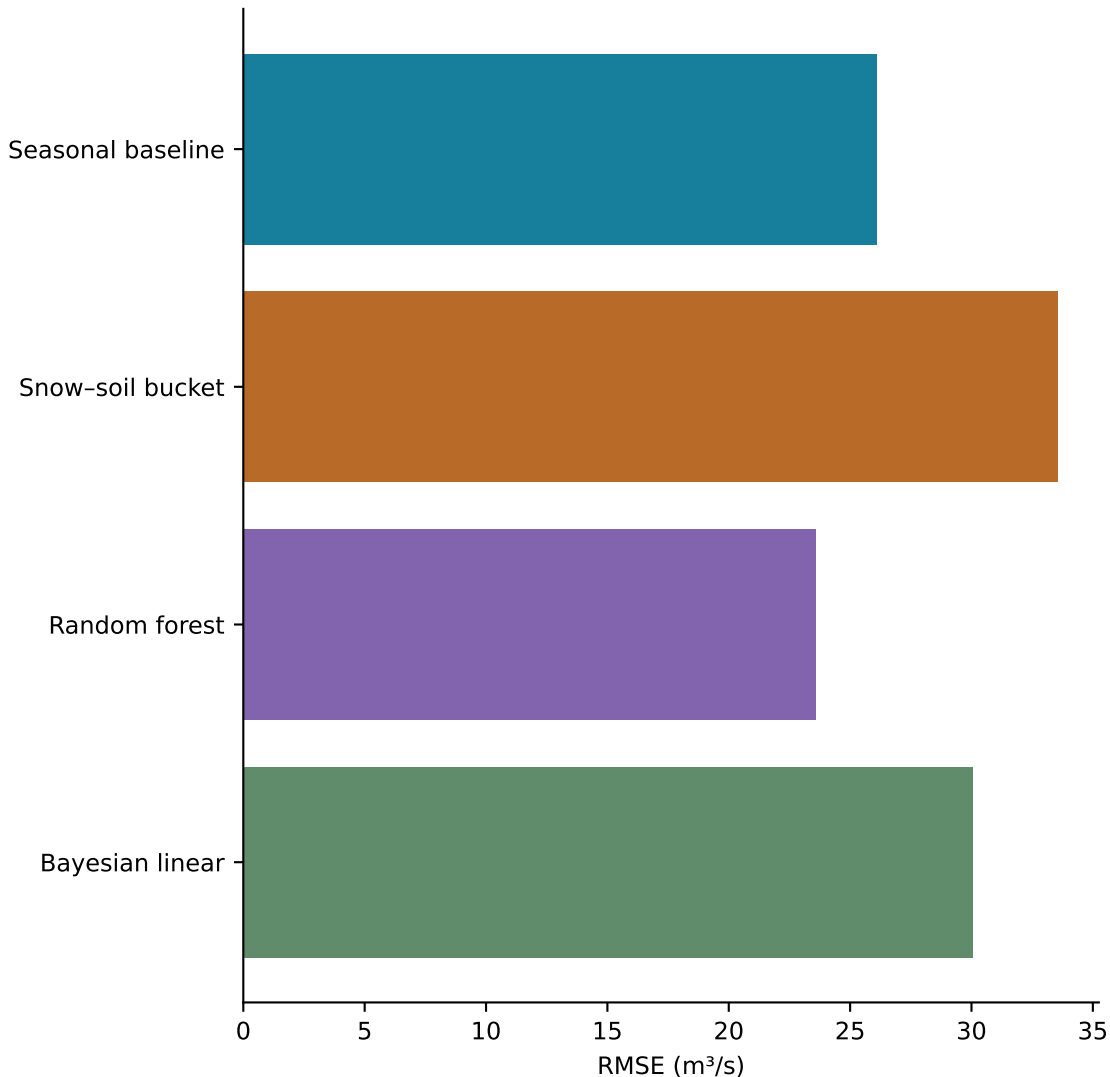
Bayesian linear • RMSE 30.1 m³/s • NSE 0.665



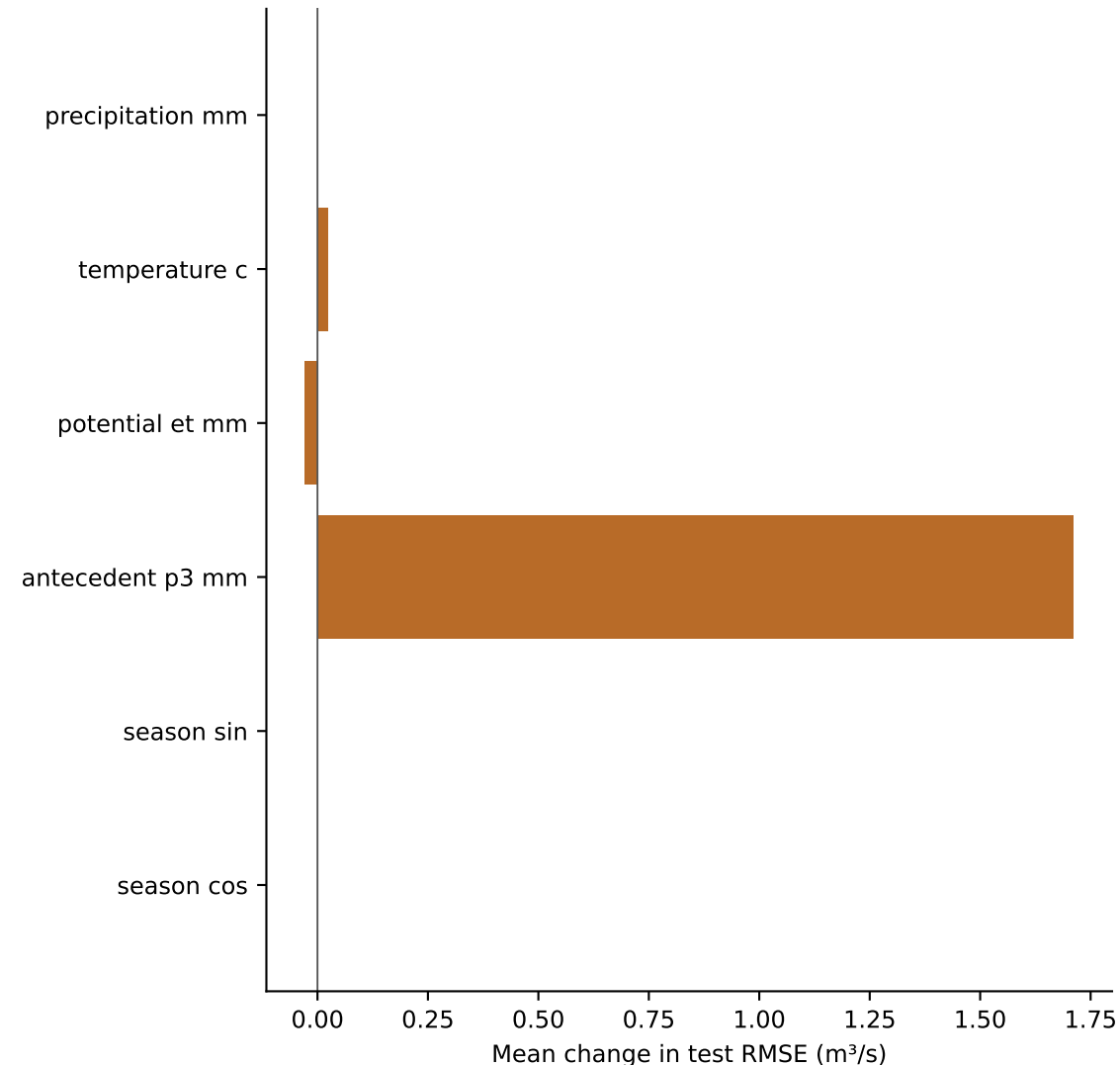
Sources: delivered screened discharge; ERA5-Land forcing. Training: 2001-2010, warm-up: 2000.
Bayesian intervals omit boundary and serial-error uncertainty. Experimental hindcasts for a provisional catchment.

Prediction skill and variable importance

a. Typical test error: lower is better



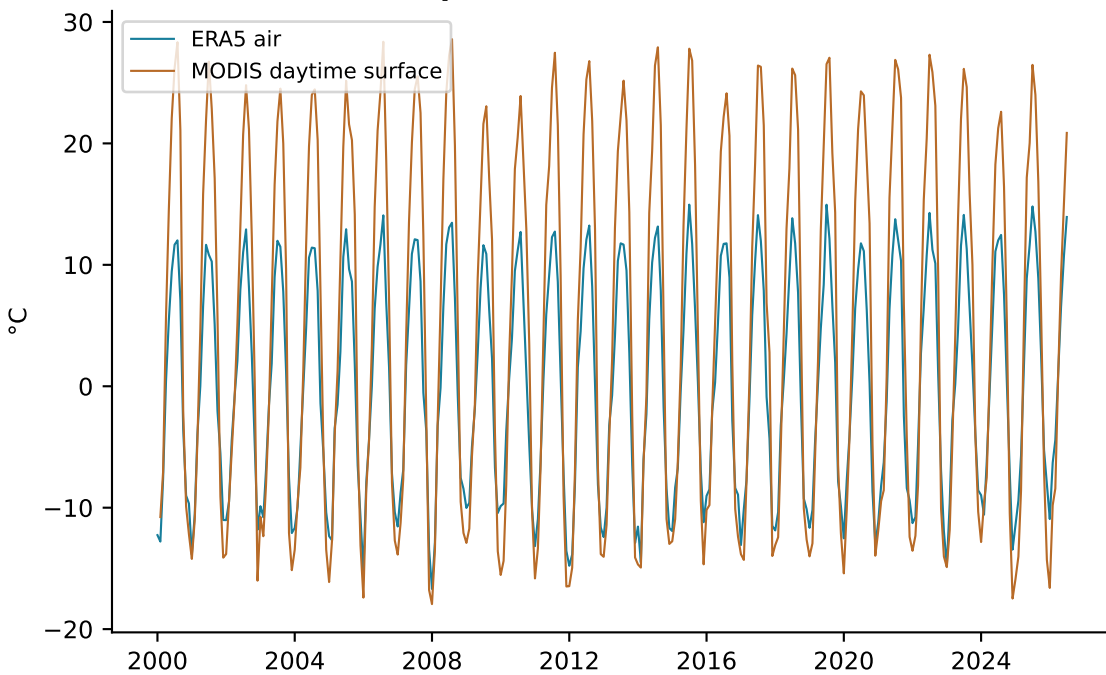
b. Forest reliance on interannual inputs



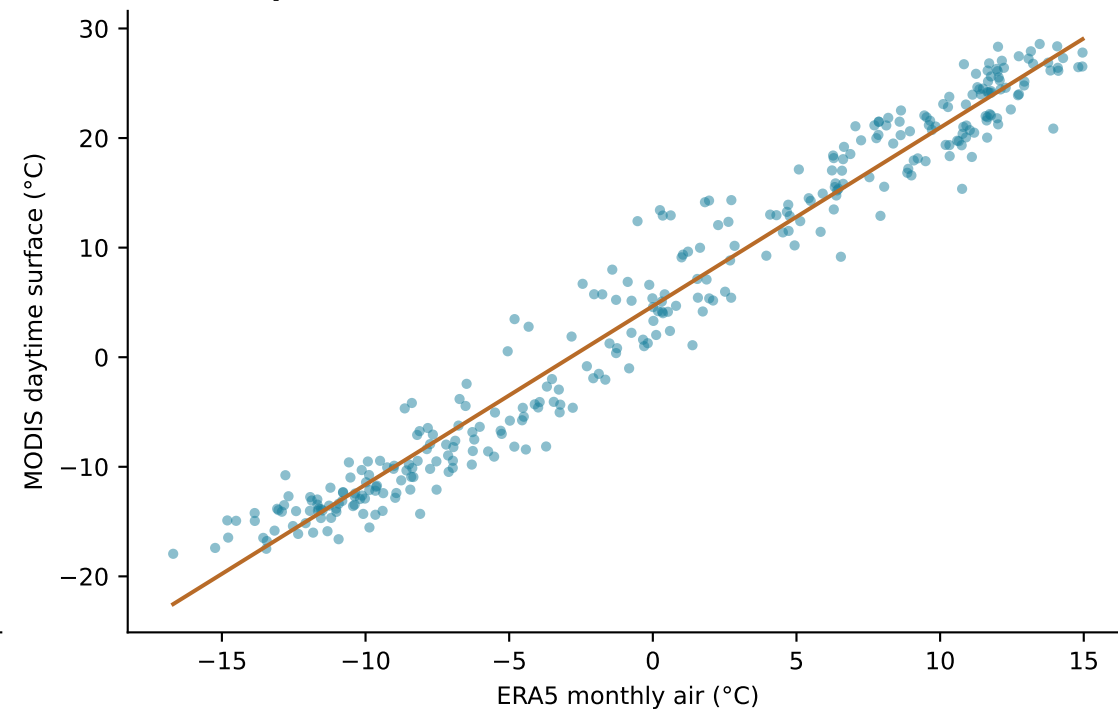
Permutation rearranges whole held-out years, preserving month alignment (20 repeats). Positive bars mean larger prediction errors.
Seasonal inputs repeat every year; zero here cannot establish that calendar season is unimportant. Association is not causation.

Pskem surface energy: seasonal relationships and different footprints

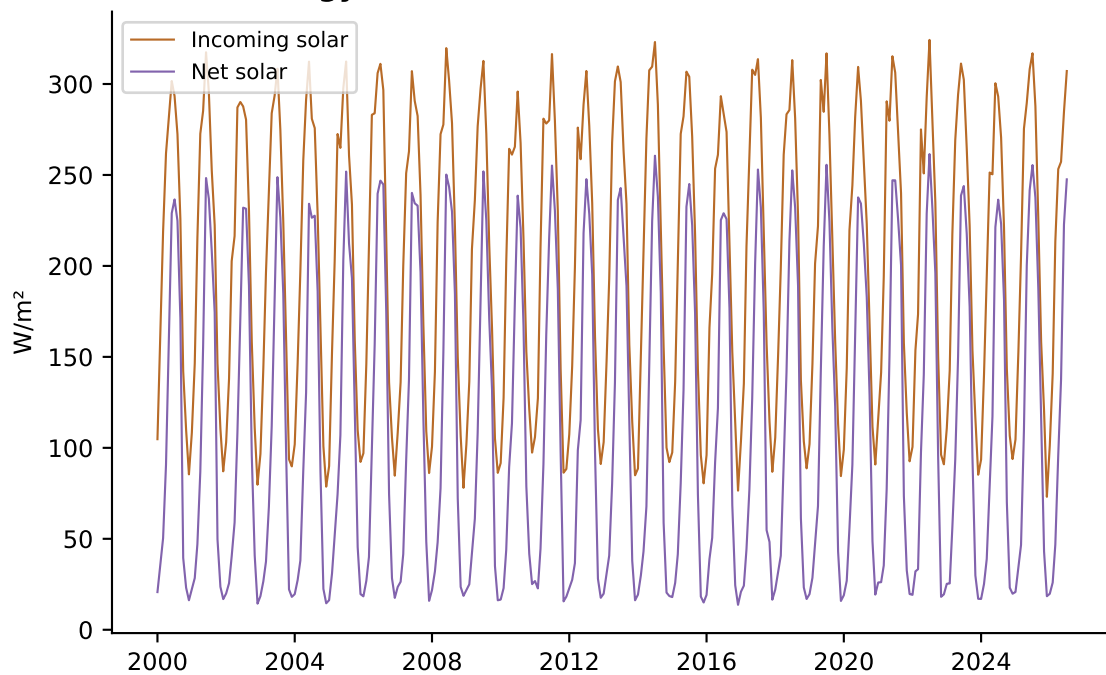
a. Two different temperatures



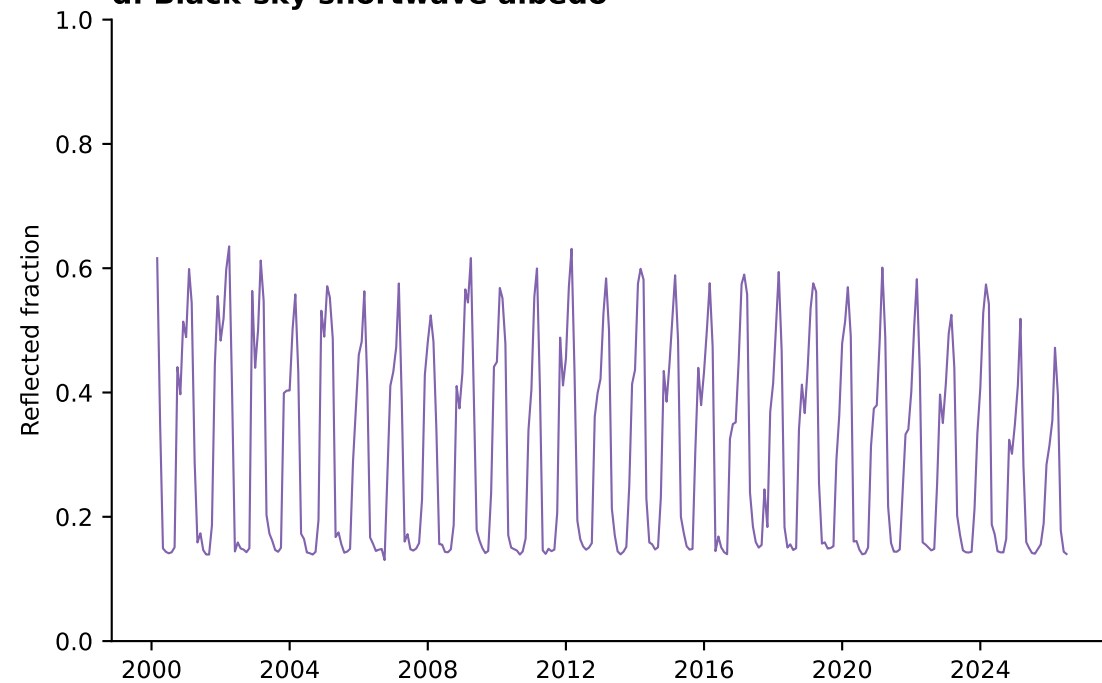
b. Descriptive fit: surface = $1.63 \times \text{air} + 4.66$



c. Solar energy



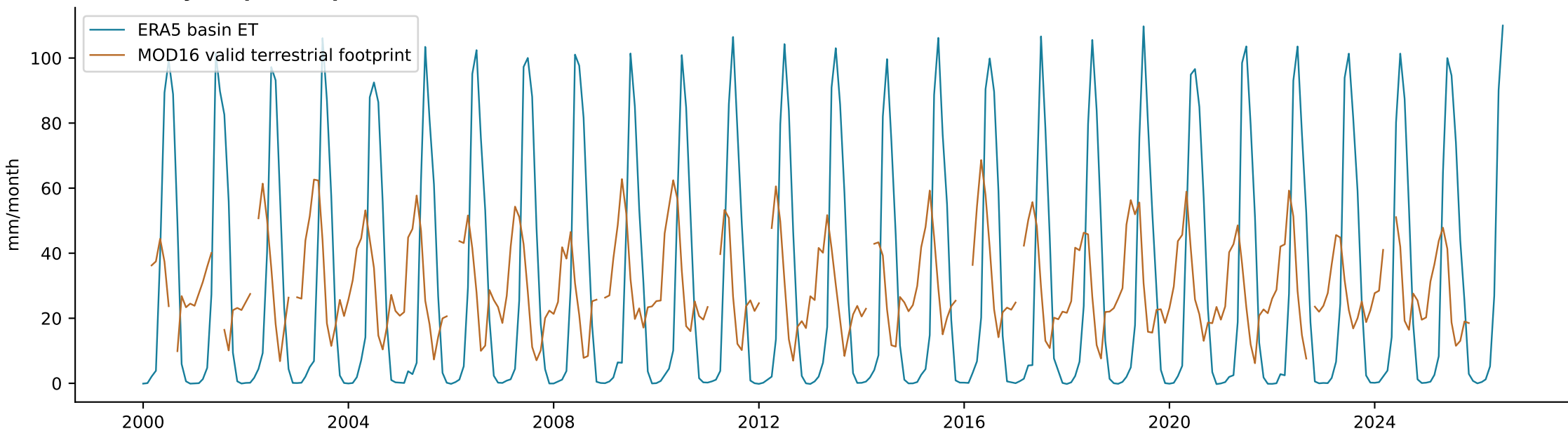
d. Black-sky shortwave albedo



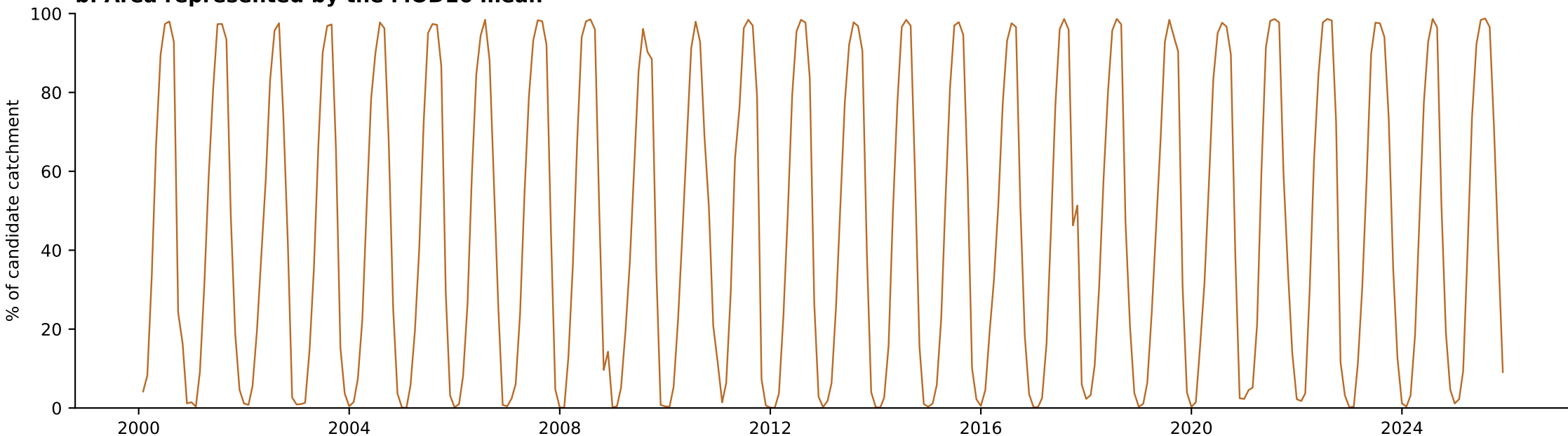
Sources: ERA5-Land, MOD11A2 and MCD43A3 v6.1. Clear-sky surface temperature is not mean air temperature.
Seasonal linear association is descriptive; a high correlation is not independent validation.

Evapotranspiration: a comparison needs matching spatial support

a. Monthly evapotranspiration

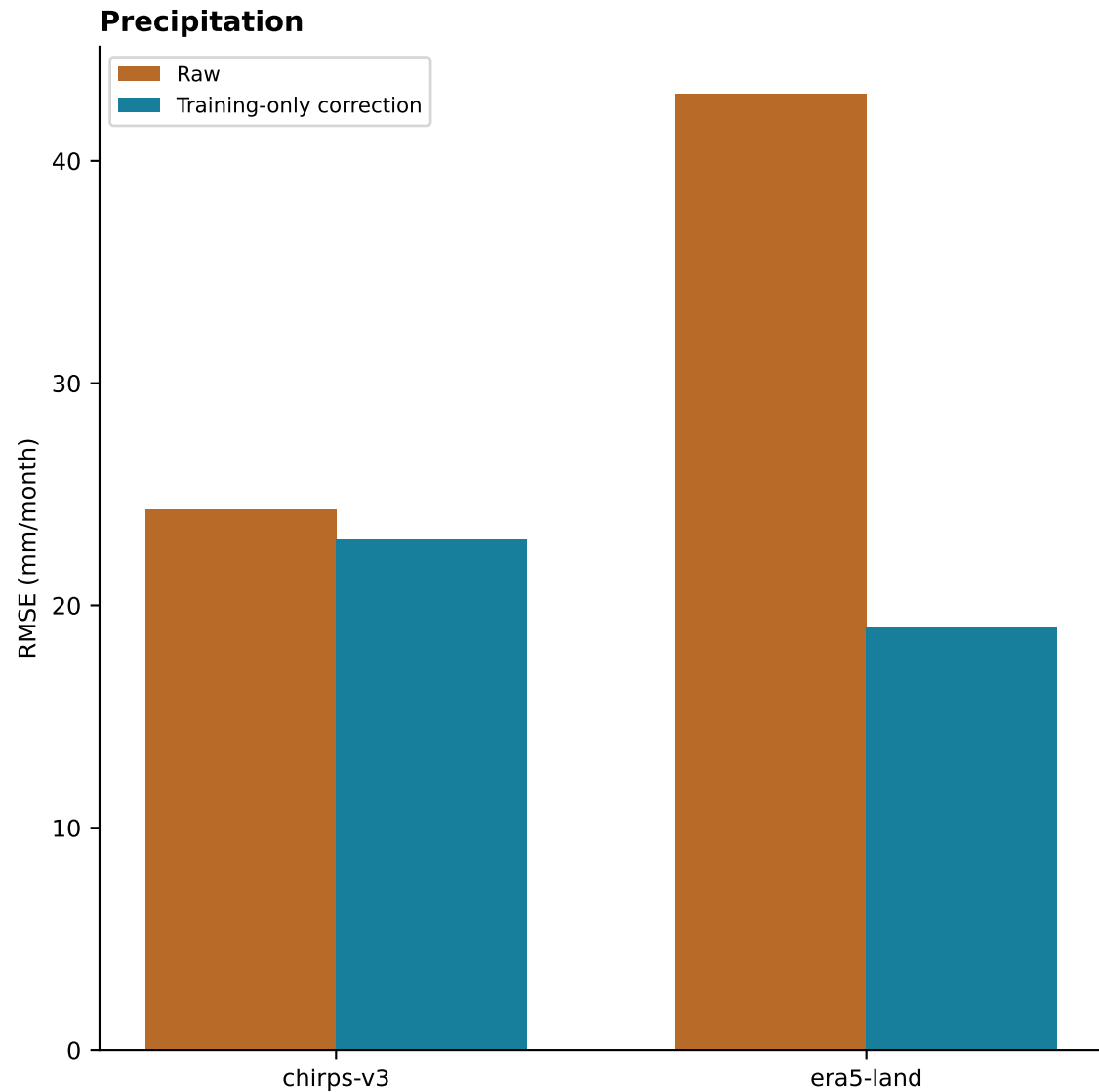
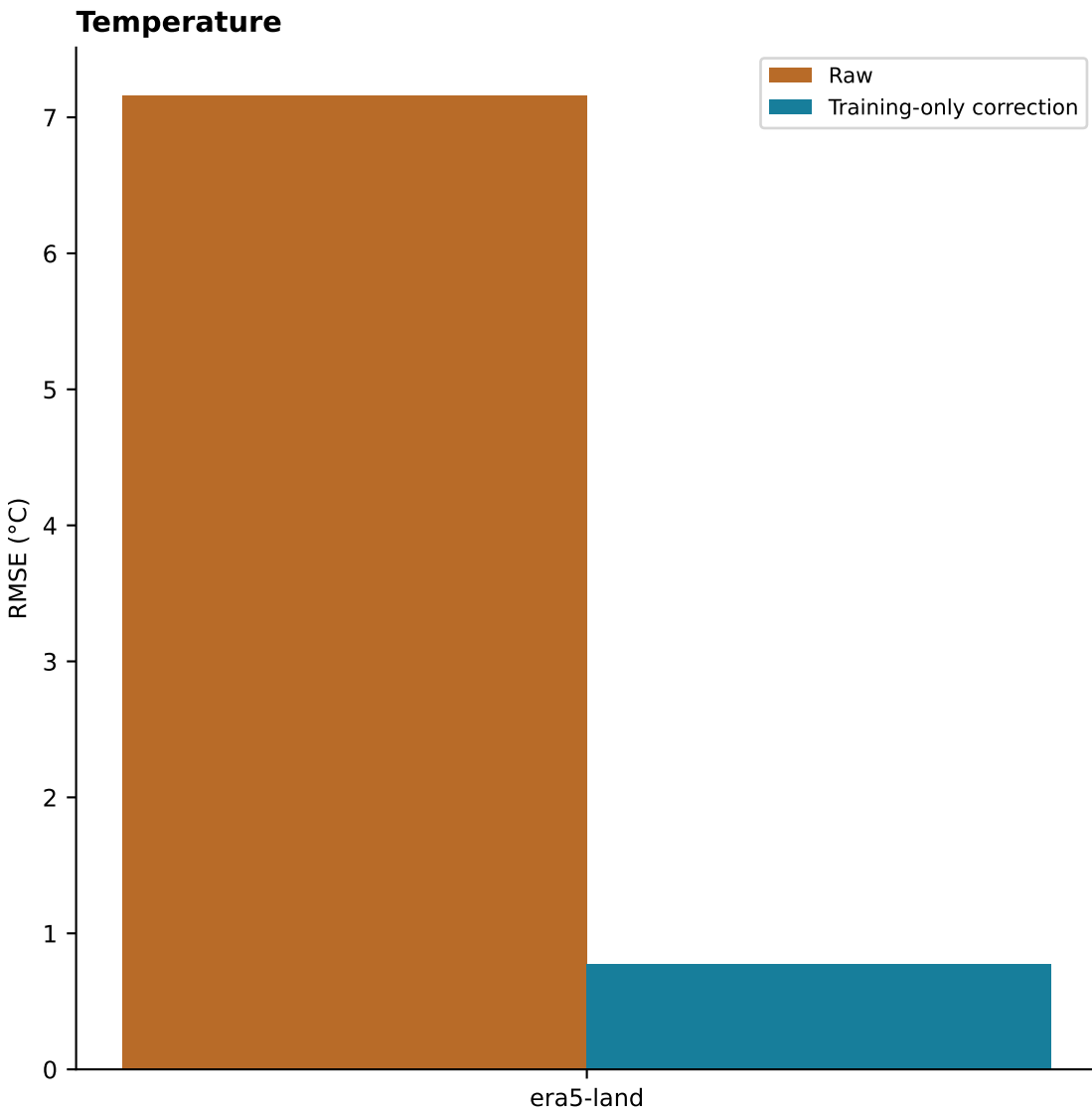


b. Area represented by the MOD16 mean



Sources: ERA5-Land and MOD16A2GF v6.1. MOD16 excludes much water, barren and snow/ice area.
Eight-day ET totals are allocated by overlapping calendar days; incomplete months remain gaps. These are model estimates, not independent ET measurements.

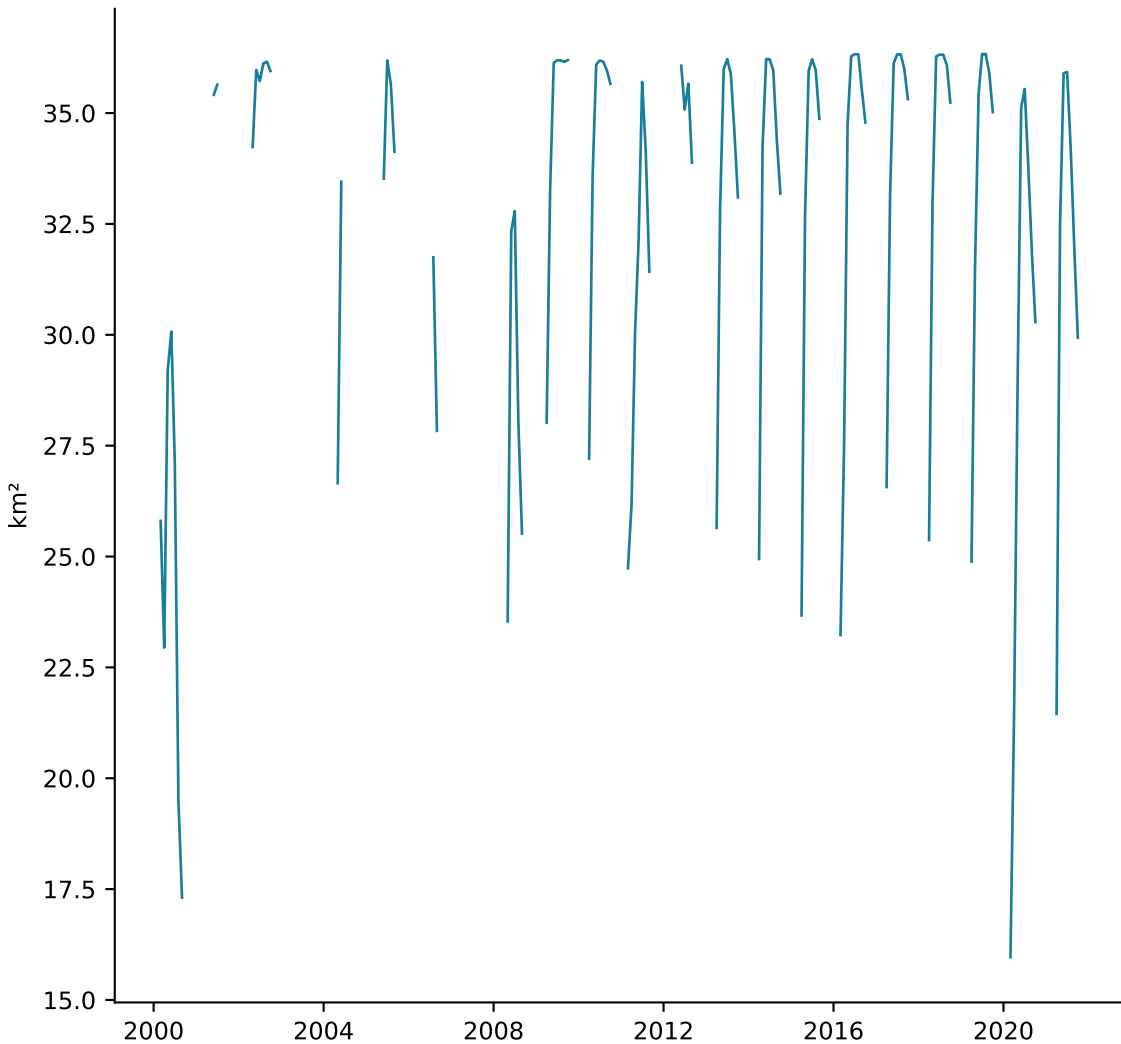
Pskem forcing correction tested on 2018-2024



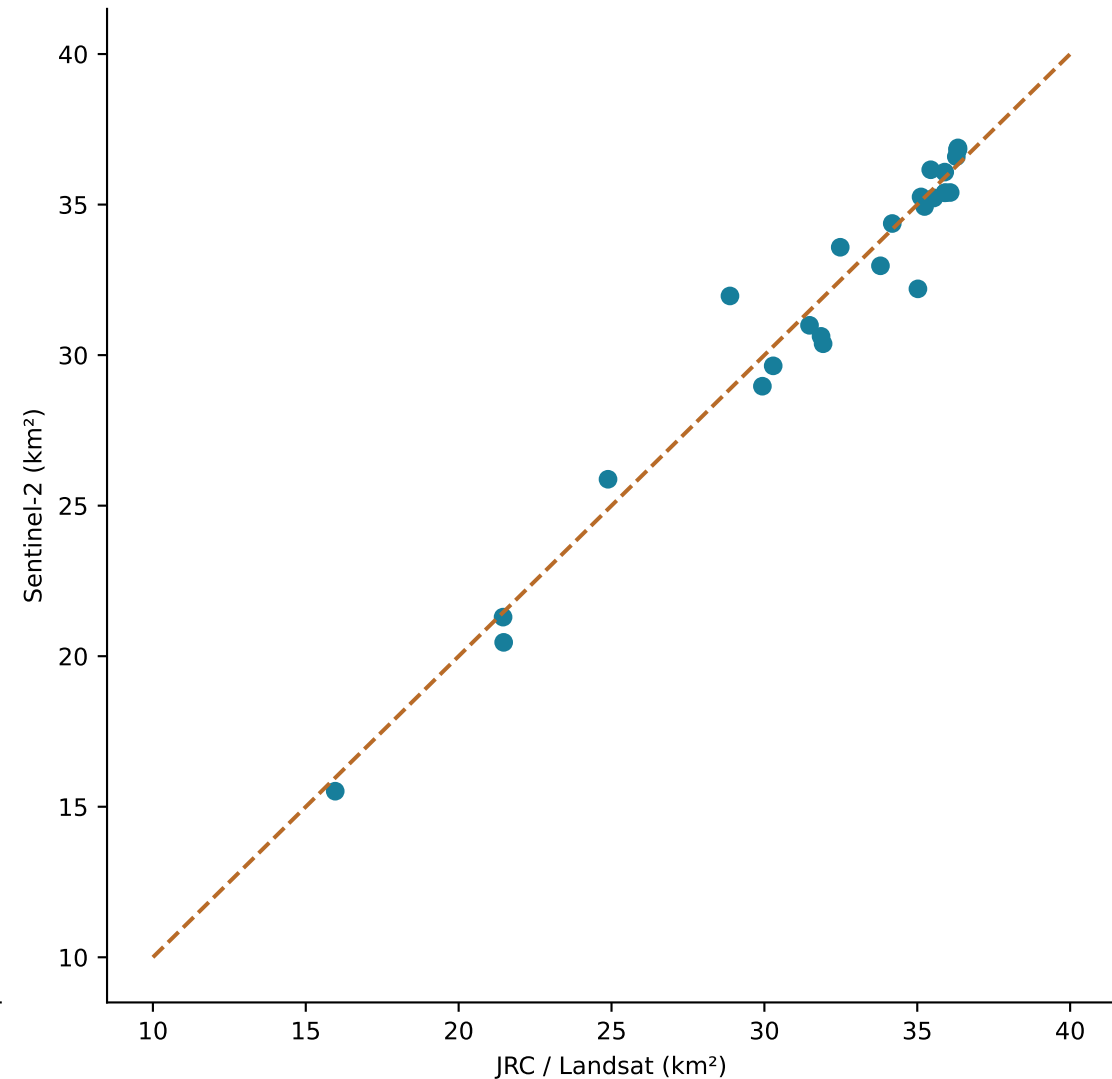
Sources: station observations, ERA5-Land, CHIRPS v3. Monthly corrections fitted only through 2017.
A successful empirical correction does not explain the original bias or validate transfer to other elevations.

Charvak: water extent and an independent-sensor check

a. Screened JRC water detections



b. 26 same-month sensor pairs



Sources: JRC Global Surface Water 1.4 and Sentinel-2 SR Harmonized. Both comparisons require $\geq 95\%$ valid area.
Acquisition dates differ within the month. The polygon plus 1 km domain includes adjacent channels; area is not storage volume.